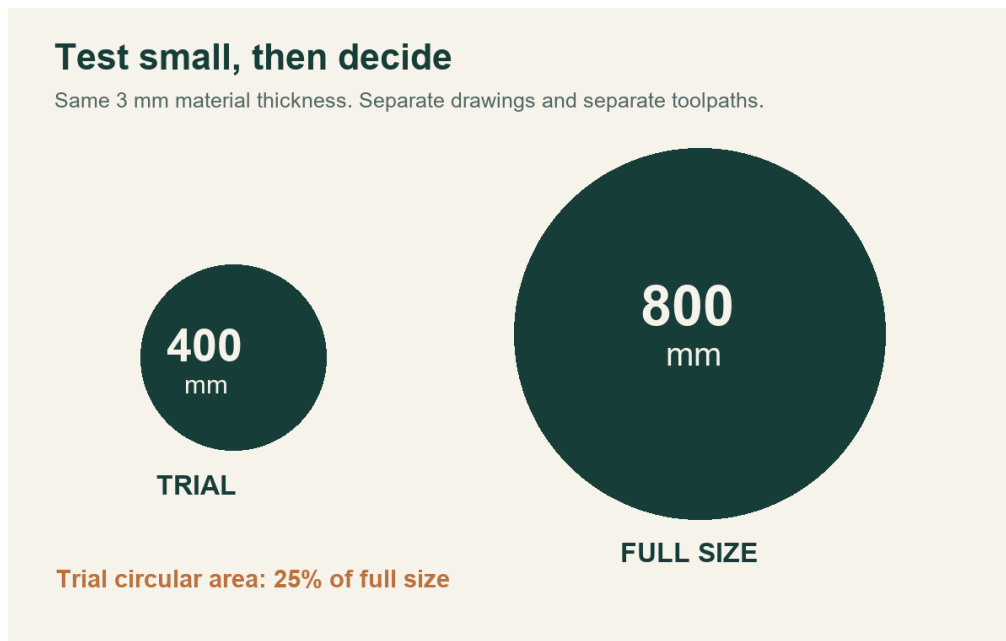


Woodlands: test cut first

An illustrated bench guide for a small complete sign, then the 800 mm version

Make the 400 mm trial in the same 3 mm mild steel planned for the final sign. Inspect it after safe removal. Cut the 800 mm sign only when you are happy with the trial and its checked cutting process.



400 mm trial first. Keep the original 800 mm artwork for the final job.

The trial contains the whole sign. Its fine details and bridges are adapted for the smaller size; it is not a simple half-scale copy. It uses 25% of the full sign's circular area. Actual stock required is larger than the sign because the complete toolpath, lead-ins and workholding need room.

Before machine work

Use the completed machine setup record from your CNC course. Machine steps are for an operator trained and authorised on this installation. If the recorded tool, post, cut values, controls or isolation procedure are missing, complete the computer preparation and obtain those details before machine work.

The controller pictures come from the supplier references already used in your CNC webpage. They show where controls are located. Their tool names, numbers and option states are examples, not your cutting recipe. Match your installed version and completed setup record.

Trial drawing: Woodlands-TRIAL-400mm.dxf

Full-size drawing: Woodlands-FULL-800mm.dxf

Prepare the trial

01 Choose the trial file

- Extract the whole guide folder from the ZIP before using its files. Open the folder named files and choose Woodlands-TRIAL-400mm.dxf.
- Keep the original 800 mm file for later. Do not shrink that file in SheetCam, and do not scale the trial's machine program up for the final cut.
- Plan a flat, identified 3 mm mild-steel offcut. A 400 x 400 mm square is only the finished design envelope; allow extra usable stock for the actual lead-ins, kerf and workholding.

Check: The selected drawing is the complete 400 mm trial, with all three name lines, foliage, two waves and two hanging holes.

Course reference for step 1

02 Put the actual machine record beside you

- Use the course's completed setup record: exact SheetCam tool and post-processor, 3 mm material recipe, measured kerf and where compensation is applied, leads/path rules, and the TAP destination folder.
- Check that the record also covers the fitted torch and consumables, physical current, air, sensing, THC/Kerf states, safe movement, dry run, stopping, cooling and electrical isolation.
- If anything is missing, you can prepare and inspect the drawing on a computer disconnected from the cutter. Obtain the missing values or demonstration before connecting or operating the machine.

Check: A usable record for this machine and this material, rather than numbers copied from a screenshot.

Course reference for step 2



The trial artwork: dark areas stay as steel; pale areas are cut away.

Import and measure

03 Import into a fresh SheetCam job

- Open SheetCam. Save any existing work, then choose File > New job.
- Choose File > Import drawing. Select Woodlands-TRIAL-400mm.dxf and click Open.
- In Drawing options, choose Metric and the bottom-left dot in the Drawing position grid, then OK. Keep the drawing at its supplied size; do not introduce a custom scale.

Check: The complete sign appears in SheetCam. Its job origin is now the bottom-left of its 400 x 400 mm bounding box, not the circle centre.

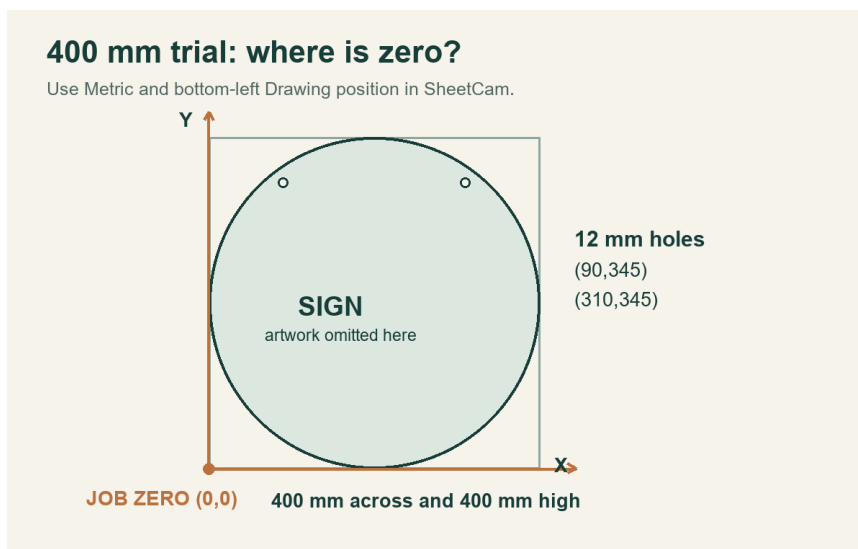
Course reference for step 3

04 Check size and position before adding toolpaths

- Choose Zoom to fit current part. Select Scroll screen, right-click the drawing and choose Measure.
- Check the circle left-to-right and top-to-bottom: both should be 400 mm. Use opposite extreme points, not a diagonal. Press Escape after measuring.
- Check the two hanging holes are 12 mm diameter. If checking centre-to-edge, that is a 6 mm radius. After bottom-left placement, the hole centres are at (90,345) and (310,345) mm.
- Compare the artwork with the supplied preview. There must be no dimension lines, a second sign, open fragments or missing letters. If measurements are unclear, verify in CAD before proceeding.

Check: A 400 mm circle, two 12 mm holes and the intended artwork. Screen appearance alone is not a size check.

Course reference for step 4



The origin is the lower-left corner of the design envelope. It is not a point on the circular rim.

Create the cutting operation

05 Create the cutting operation

- Under Operations, click the jet-cutting icon to open Jet cutting operation.
- Use the course's Outside Offset whole-part method and select layer CUT, which contains the supplied closed contours. Select the exact tool from the completed record for this material and thickness.
- Check Feed rate, Path rules and lead-in/out type and length against that record. Arrange starts in waste material; a lead must fit its opening without crossing retained lettering or a bridge.
- Click OK. Never copy the example tool, feed, material thickness or lead length shown in the supplier picture.

Check: The operation uses CUT and the correct recorded tool. Every internal opening has a waste-side path; the perimeter path lies outside the sign.

Course reference for step 5

Jet cutting

Basic | Cut path | Notes

Contour method: Outside Offset

Layer: 0

Tool: T7: 3mm Mild Steel 45A

Feed rate: 3444 mm/min

Max chain length: 0 mm

Overcut: 2 mm

Offset open paths: ☐

Leadins on open paths: ☐

Reverse cut direction: ☐

Path rules: My Rule Set

Lead in:

- ☐ None
- ☐ Arc
- ☐ Tangent
- ☒ Perpendicular

Length: 5 mm

Lead out:

- ☒ None
- ☐ Arc
- ☐ Tangent
- ☐ Perpendicular

Length: 0 mm

Start at the centre of circles smaller than: 40 mm

Use code snippet: None

Loop sharp corners

Loop size: 0.6 mm

☒ None

☐ Triangle

☐ Arc

Angle threshold: 0 degrees

Overcut internal corners

Angle threshold: 0 degrees

Overcut scale: 200 %

☐ Enabled

Material thickness: 10 mm

Edit material

OK Cancel Help

EXAMPLE SCREEN ONLY. Locate the controls; use your own recorded tool and values. The imported layer is CUT, not the pictured layer 0.

Check every path and make the TAP

06 Watch the whole preview, then make the TAP

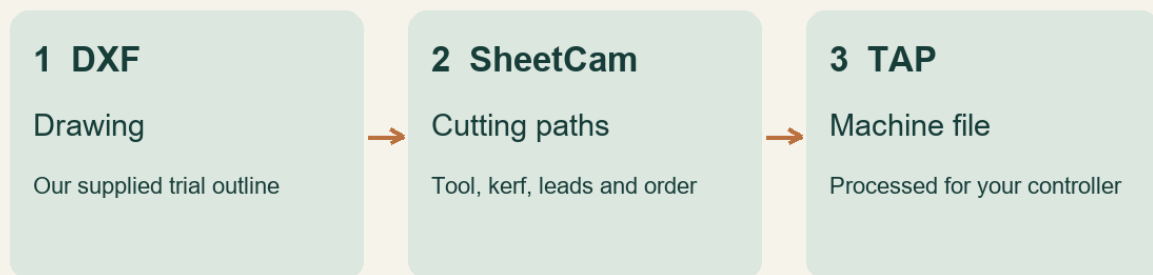
- Use Show input paths, Show tool paths, Show path directions, Show cut start points and Show rapid moves. Inspect the small letters and bridges closely.
- Run simulation in SheetCam. This is a screen preview. Check the trial's 38 closed contours: 35 artwork openings and two mounting holes first, then the outside circle last. Each contour must cut once. Check all paths and leads fit the usable offcut and miss clamps.
- If a cut is missing, doubled, open or on the wrong side, correct the drawing/operation and replay the whole preview. Do not add a second operation blindly.
- Choose File > Save job as and save a clearly named trial job. Verify the supplier-confirmed post, then File > Run post processor. Save Woodlands_TRIAL_400_v1.tap. Resolve any processing error before continuing.
- Copy that exact TAP revision to the machine-computer folder in the record. Keep the DXF, editable SheetCam job and TAP together. Renaming a DXF to .tap does not convert it.

Check: The exact checked trial TAP is in the correct folder. The job was previewed all the way through, with internal cuts first and the circle last.

Course reference for step 6

The file journey

Prepare the toolpaths in SheetCam before loading SureControl.



Load the checked TAP in SureControl

Then set origin, dry run and cut.

DXF is the drawing. SheetCam creates cutting paths. TAP is the file SureControl runs.

Prepare the machine

07 Load the offcut using the recorded procedure

- Complete the machine's pre-use checks: clear dry area, flammables removed, serviceable guarding, screens, extraction, access controls and emergency arrangements.
- Keep the required electrical isolation in place for loading. Use the recorded lifting/handling protection. Support and secure the flat 3 mm offcut so the cut and waste pieces will not cause a collision.
- Match the actual material, torch, consumables, current, air and work-return connection to the setup record. Keep clamps and the complete cut envelope clear of travel.
- Follow the recorded power-on sequence. Everyone remains outside the controlled travel area; start extraction at the specified point before any firing.

Check: Identified, secure stock with enough usable area for the complete trial toolpath and leads, and a physical setup matching the job record.

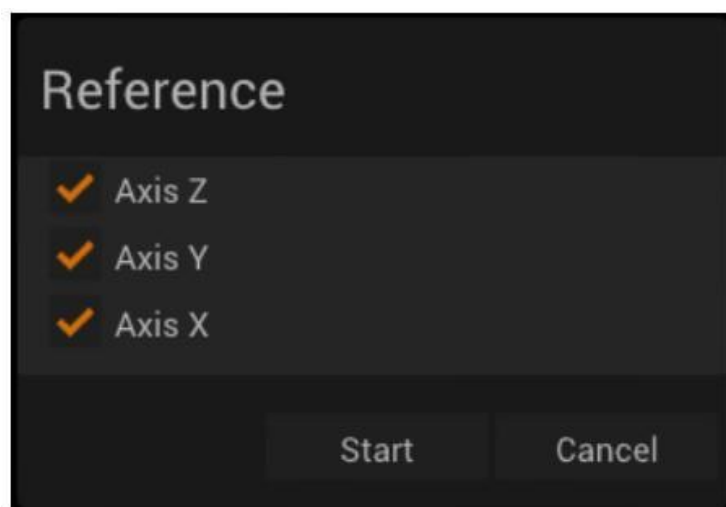
Course reference for step 7

08 Connect and reference SureControl

- Open SureControl. In the documented fresh-start sequence it initially reads Disconnected. Click Connect; the expected state is Offline and Connect changes to Reset. Do not press Reset just because the label changed.
- With the prepared system clear of people, enable Online. The documented fresh start then expects Not Referenced. Resolve any unexplained state or fault before continuing.
- Click Reference. Use the axes and sequence confirmed for this installation. Click Start only with the travel area clear, wait for normal completion, then OK.
- Referencing establishes machine positions; it does not position the sign on the sheet.

Check: The required axes are referenced, with no unexpected motion or unresolved fault. Offline is not electrical isolation.

Course reference for step 8



Supplier Reference dialog. Start causes real machine movement; use the recorded axis sequence.

Load and check

09 Load the exact trial TAP

- Click Load. In Load Program, use Directories to find the confirmed folder and select Woodlands_TRIAL_400_v1.tap.
- Leave Import DXF unticked: the cutting paths have already been prepared in SheetCam. Click OK.
- Check both the filename and complete sign preview against your trial job. Confirm the expected 400 mm job and orientation. Do not press Start yet.

Check: The correct trial program is loaded. A fresh unpositioned job may show Set Origin, as described in the supplier reference.

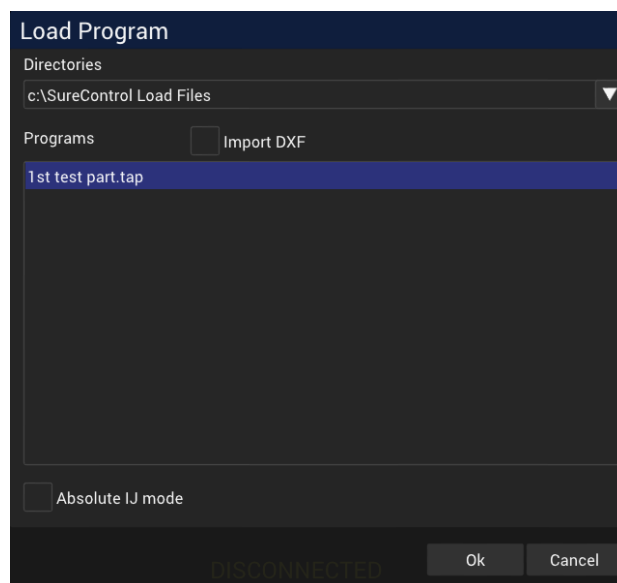
Course reference for step 9

10 Check sensing and the recorded torch test

- Perform the Touch sensor test recorded for the fitted sensing system. Touch in Indicators shows the response; Touch in Options enables the function. They are different controls.
- For the manual's contact-test method only where confirmed for this machine: use the recorded low jog speed and clearance, Page Down gently to sensing contact, release immediately, then Page Up to the recorded clearance. Check the indication changes and clears. Do not hold Shift or force the torch into the sheet.
- Follow the record on whether a torch test is required. If required, use its recorded position, clearance, duration, screens and protection. Click Torch On/Off once to begin and click the same icon again to stop at the recorded point; verify the arc is out. A blank test entry is not a reason to experiment.

Check: Sensing responds normally, the torch is safely clear, and the specified test is complete or recorded as unnecessary. Resolve a fault before going further.

Course reference for step 10



Supplier Load Program reference: load the checked TAP, with Import DXF unticked.

Position and dry run

11 Set the lower-left job origin

- Identify the lower-left corner of the 400 x 400 mm design envelope used in SheetCam. Choose a position that leaves space for all kerf, lead-ins and travel as well as the sign.
- At the recorded safe clearance and low jog speed, use the recorded X/Y jogging controls to reach that position. For the documented keyboard method, arrows move X/Y while held; release to stop. Do not hold Shift.
- Release the keys, then click Set Origin. Do not click Origin, which commands movement to a previously assigned zero.
- Check the part coordinates and orange crosshairs match the intended job zero. The next step verifies the real travel.

Check: The origin is assigned to the planned lower-left design-envelope corner. The 400 mm circle fits the offcut with space for the complete toolpath.

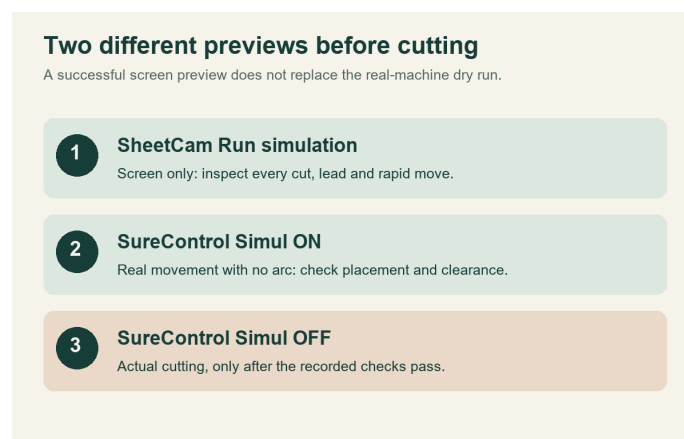
Course reference for step 11

12 Run the complete dry run

- Use the dry-run conditions and clearance in the setup record. Confirm the travel area is clear, Online is enabled and Simul is ON/highlighted.
- Click Start from the beginning of the program. If the start-menu wording is unclear, cancel and have the trained operator identify it; do not choose a restart or nearest-profile option for this fresh job.
- Watch the whole sequence from the screened operator position. The machine moves, but the torch must not fire. Check every internal path, the outside circle, leads and travel stay clear of edges and clamps.
- Wait for completion and Idle. Leave Simul ON while checking. If the program or origin changes, repeat the complete dry run.

Check: The full trial route completes without firing, obstruction or unexpected travel. SureControl Simul moves the actual machine; it is different from SheetCam's screen preview.

Course reference for step 12



Screen preview first, real-machine dry run second, actual cutting only after both checks pass.

Make the trial cut

13 Make the trial cut

- Recheck the filename, origin, actual 3 mm sheet, torch/consumables and physical current against the record. Any change since the dry run requires the checks to be repeated.
- Open Cut Parameters using the sliders icon and compare the relevant values with the completed record. Confirm which values the TAP supplies and which SureControl applies. Do not overwrite CAM-controlled settings or apply kerf compensation twice.
- Read the Options states against the record. The documented plasma setup has Touch ON and Laser OFF; THC and Kerf must have the individually recorded states. Confirm Online ON. Deliberately turn Simul OFF only when ready for the real cut.
- With extraction running and access secured, remain at the screened controls and start from the beginning. Observe the whole cut. For faults, lost extraction or abnormal behaviour, use the recorded stop/emergency procedure; do not improvise a restart.

Check: All internal openings are cut before the perimeter. The trial cut finishes under the recorded machine conditions.

Course reference for step 13

Cut Parameters

65 Amp

Carbon steel

Quality

Del

New

Save

3.0	mm	Plate thickness	10.000
4.0	mm	Cutting speed	1040.0
6.0	mm	Volts	137.0
8.0	mm		
10.0	mm	Pierce height	3.8
12.0	mm	Pierce delay	0.7
16.0	mm	Cutting height	3.2
		Kerf	1.9

Optimise touch

Touch distance

50 mm

Ok

Cancel

EXAMPLE SCREEN ONLY. Check Cut Parameters against the installed machine's completed record; do not copy these numbers.

Retrieve and inspect

14 Finish, isolate and remove safely

- Wait for motion to stop, the arc to be out and the controller to report Idle. Keep access controlled; Idle is not electrical isolation.
- Follow the recorded shutdown, cooling/post-flow and extraction run-on sequence. Apply and verify electrical isolation using the established method before access or retrieval.
- Use the recorded cooling and handling method, tools and protection. Treat the sign, waste and remaining sheet as hot and sharp. Do not touch the metal to test its temperature or reach into an energised table.

Check: Safe isolated equipment and a trial sign retrieved under the recorded handling conditions.

Course reference for step 14

15 Decide whether you are happy with the trial

- After safe cleaning, check every opening has cut through. All letter centres, bridges and gum-leaf details should remain attached; no melted-through connections or partly detached details.
- Measure the circle in two directions against 400 mm and the hanging holes against 12 mm. Record actual values and compare with the tolerance agreed for your machine and hardware.
- Look for unacceptable bowing, excessive dross, bevel, gouged starts or rough lettering. Stand back and judge whether the words, foliage and overall appearance work for you.
- Record the tool/recipe, TAP revision, measurements and a photo. If anything is unsatisfactory, stop the full-size job. Have the cause corrected, regenerate a new trial TAP and repeat the preview and dry run before retesting.

Check: A deliberate decision: HAPPY TO PROCEED or REVISE AND RETEST. Finishing paint will not repair a weak bridge or incomplete cut.

Course reference for step 15

Inspect before committing the full sheet

Make a deliberate decision after safe cooling, isolation and retrieval.

Cut-through Every opening is complete	Connections No failed bridge or loose detail
Dimensions 400 mm sign; 12 mm holes	Finish Acceptable dross and bevel
Flatness No unacceptable bowing	Appearance Words and foliage look right

HAPPY TO PROCEED / REVISE AND RETEST

Check attachment, cut-through, dimensions, flatness and appearance before using the full sheet.

Prepare the full-size sign

16 Prepare the original 800 mm sign as a new job

- Only if you are happy with the trial, start a fresh SheetCam job and import Woodlands-FULL-800mm.dxf. Use Metric and the same bottom-left drawing-position method.
- Confirm 800 mm diameter in both directions, two 12 mm holes and 39 closed contours in total: 36 artwork openings, two mounting holes and the outside circle. Do not enlarge the trial DXF or TAP; use this original full-size drawing.
- Use the checked material/tool setup, then rebuild and review leads, path rules, heat distribution and cut order for this larger job. Save a separate job and Woodlands_FULL_800_v1.tap.
- Repeat the complete machine setup, file/origin checks and full dry run on the actual larger sheet. The trial supports your decision but does not prove the 800 mm job's placement, heat behaviour or toolpaths.
- Cut, isolate, retrieve and inspect using the same checked sequence. Keep both trial and full-size records.

Check: A separately checked full-size job, cut only after your trial decision and the full-sheet checks are satisfactory.

Course reference for step 16

Test small, then decide

Same 3 mm material thickness. Separate drawings and separate toolpaths.



TRIAL



FULL SIZE

Trial circular area: 25% of full size

400 mm trial: one-quarter circular area. 800 mm final: original artwork and its own toolpaths.

Trial record

Record the result before choosing the full-size cut.

Name / date: _____

Material / measured thickness

Exact SheetCam tool / recipe

Confirmed post-processor

Kerf width / where applied

Lead and path-rule record

TAP filename / revision

Measured diameter X / Y

Measured hole diameters

Cut-through / bridges / flatness

Decision: proceed or revise

Course references

Your CNC guided operating sequence

Machine setup record

Your settings reference

SheetCam review in your CAD guide

Keep the cut photo, drawing, editable SheetCam job and exact TAP revision together.